

ANCHORAGE BIRCHWOOD, AK, USA

WMO: 702746

Lat: 61.416N

Lon: 149.507W

Elev: 96

StdP: 14.64

Time Zone: -9.00 (NAK)

Period: 95-19

WBAN: 26497

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.5	-7.9	-21.5	1.7	-9.0	-15.3	2.4	-5.1	24.7	23.9	18.7	33.8	1.1	150	0.638

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.1	73.1	59.6	70.3	58.1	66.8	56.3	61.2	71.4	59.2	67.7	57.6	65.0	5.4	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
56.4	68.0	64.3	54.8	64.1	62.4	53.9	62.1	61.5	27.1	71.4	25.7	67.7	24.8	65.0	72.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.7	11.8	9.7	-15.9	77.7	5.3	4.2	-19.7	80.8	-22.8	83.2	-25.8	85.6	-29.6	88.7		
(5)			-16.3	63.0	4.7	3.0	-19.6	65.1	-22.3	66.8	-25.0	68.5	-28.3	70.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.7	15.2	19.8	24.7	37.7	48.1	56.1	58.9	56.3	48.4	35.4	21.7	16.6		
	DBStd	17.93	12.93	10.37	9.35	5.80	5.15	4.15	3.58	3.84	5.02	8.27	10.54	11.55		
	HDD50	5604	1079	846	786	370	96	2	0	2	83	455	848	1037		
	HDD65	10351	1544	1266	1251	819	523	269	192	271	497	919	1298	1502		
	CDD50	734	0	0	0	1	38	184	277	197	36	1	0	0		
	CDD65	7	0	0	0	0	0	2	4	1	0	0	0	0		
	CDH74	74	0	0	0	0	1	23	39	11	0	0	0	0		
	CDH80	9	0	0	0	0	0	2	7	0	0	0	0	0		
	WSAvg	3.2	3.0	2.9	3.6	3.3	4.2	3.5	2.8	2.6	2.8	3.2	3.2	3.3		
	PrecAvg	18.0	1.1	1.0	0.8	0.5	0.7	1.1	1.9	2.8	3.3	1.9	1.4	1.5		
(15) Precipitation	PrecMax	23.3	2.2	2.4	2.0	1.8	1.7	2.6	3.6	6.7	6.9	4.6	3.0	2.7		
	PrecMin	13.3	0.2	0.1	0.1	0.1	0.2	0.3	0.6	1.1	1.0	0.7	0.1	0.1		
	PrecStd	2.4	0.6	0.6	0.4	0.4	0.4	0.6	0.8	1.4	1.6	0.9	0.7	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.9	44.8	46.0	60.9	72.2	77.3	79.1	75.3	64.3	55.2	47.7	45.7		
		MCWB	37.7	37.5	38.6	47.4	54.8	60.7	62.8	61.4	53.1	47.1	40.6	38.8		
	2%	DB	39.2	39.0	42.4	54.3	66.2	72.2	73.1	72.1	61.5	51.6	41.5	39.2		
		MCWB	35.1	34.9	36.1	43.2	52.1	57.8	60.5	59.9	51.8	45.2	37.5	35.4		
	5%	DB	36.2	36.4	39.0	51.6	62.6	68.5	70.2	68.0	58.9	48.1	37.3	35.6		
		MCWB	33.6	33.6	33.8	41.4	49.8	56.4	59.0	57.8	51.1	43.6	34.6	33.2		
	10%	DB	32.7	33.9	36.8	47.9	58.5	64.9	66.4	64.2	56.5	45.8	35.5	31.5		
		MCWB	30.9	31.6	32.5	39.9	47.8	54.3	57.1	55.3	50.0	41.9	33.6	30.6		
	0.4%	WB	38.2	38.4	39.9	48.3	56.2	62.3	64.4	63.2	55.6	49.6	41.1	39.4		
		MCDB	42.2	42.9	44.4	59.8	69.8	74.0	76.5	72.9	60.7	53.4	46.5	45.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	36.2	35.5	37.0	44.1	52.8	59.1	61.4	60.7	53.9	46.3	38.1	35.8		
		MCDB	39.4	38.3	40.9	52.2	64.2	69.8	71.8	69.9	58.4	49.0	41.1	39.3		
	5%	WB	33.3	33.6	34.8	41.9	50.8	57.1	59.5	58.5	52.6	44.4	35.5	32.7		
		MCDB	35.8	36.5	38.5	49.2	60.5	66.5	67.7	66.1	57.0	47.2	37.3	34.7		
	10%	WB	30.5	31.2	32.8	40.3	48.9	55.2	58.1	56.7	51.2	42.4	32.9	30.1		
		MCDB	32.3	33.3	36.4	47.3	57.6	63.5	65.4	63.2	55.6	45.7	34.7	31.2		
	5% DB	MDBR	11.8	13.9	16.7	16.5	16.3	14.6	13.1	14.0	13.6	12.0	11.2	11.8		
		MCDBR	13.0	13.6	15.0	21.0	22.7	21.4	20.2	21.0	17.4	13.4	12.1	15.1		
		MCWBR	9.8	10.3	10.4	12.5	13.1	11.2	11.0	11.4	10.2	8.7	9.3	11.6		
		MCWBR	12.0	13.1	14.2	19.1	19.9	18.3	17.2	18.1	14.3	11.2	11.7	14.1		
(40) Clear-Sky Solar Irradiance	taub		0.233	0.286	0.306	0.343	0.364	0.362	0.371	0.353	0.314	0.296	0.243	0.383		
		taud	2.114	2.222	2.259	2.249	2.288	2.380	2.394	2.453	2.546	2.425	2.099	2.659		
	Ebn at Noon		156	218	257	268	271	274	268	264	258	218	144	94		
		Edh at Noon	11	21	28	35	36	34	33	29	22	17	11	8		
(44) All-Sky Solar Radiation	RadAvg		90	311	823	1371	1645	1694	1515	1202	778	357	131	47		
	RadStd		12	37	79	113	128	104	114	121	99	40	18	6		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (25 neighbors)		N/A	+3.56	+2.95	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page